

Structural Vibration Analysis Update Multi-Sensor Accelerometer Measurements on a 3D Printer

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1 Objective

The purpose of this experiment was to continue evaluating the feasibility of using low-cost MEMS accelerometers for structural vibration analysis applications. A three-sensor configuration was implemented to observe vibration transmission through different regions of the printer structure and to compare the dynamic response at each location.

This report serves as a progress update of the experimental setup, data acquisition process, and preliminary vibration analysis results obtained from the printer structure.

2 Experimental Setup

Three MPU6050 accelerometers were used simultaneously during the test. All sensors were mounted using rigid screw-mounted brackets directly attached to the printer structure in order to minimize mounting flexibility and measurement damping caused by adhesive or soft mounts.

The sensor locations were defined as follows:

- **Sensor 1 (s1):** Mounted at the base of the printer frame.
- **Sensor 2 (s2):** Mounted on the opposite side of the structure toward the upper section of the printer frame.
- **Sensor 3 (s3):** Mounted directly on the printer head/extruder assembly.

The use of multiple sensors allowed the observation of vibration propagation through the structure and enabled comparison between the rigid frame response and the motion occurring at the moving print head assembly.

3 Data Acquisition Parameters

The vibration data was collected using an ESP32-based acquisition system operating at a high sampling rate.

Table 1: Data Acquisition Parameters

Parameter	Value
Estimated Sampling Rate	800 Hz
Sampling Interval	0.00125 s
Total Duration	16.714 s
Number of Samples	13,372
Nyquist Frequency	400 Hz
Data Mode	Three Sensor Mode

The sampling rate was selected to allow observation of structural frequencies up to approximately 400 Hz while maintaining sufficient temporal resolution for transient vibration events.

4 Time Domain Results

Figures 1, 2, and 3 show the acceleration response measured by each sensor in the time domain.

A transient excitation event can be observed near the 5-second mark in all sensors. The response shows a damped oscillatory behavior following the excitation, indicating structural vibration decay after the impulse event.

4.1 Sensor 1 – Base of Printer

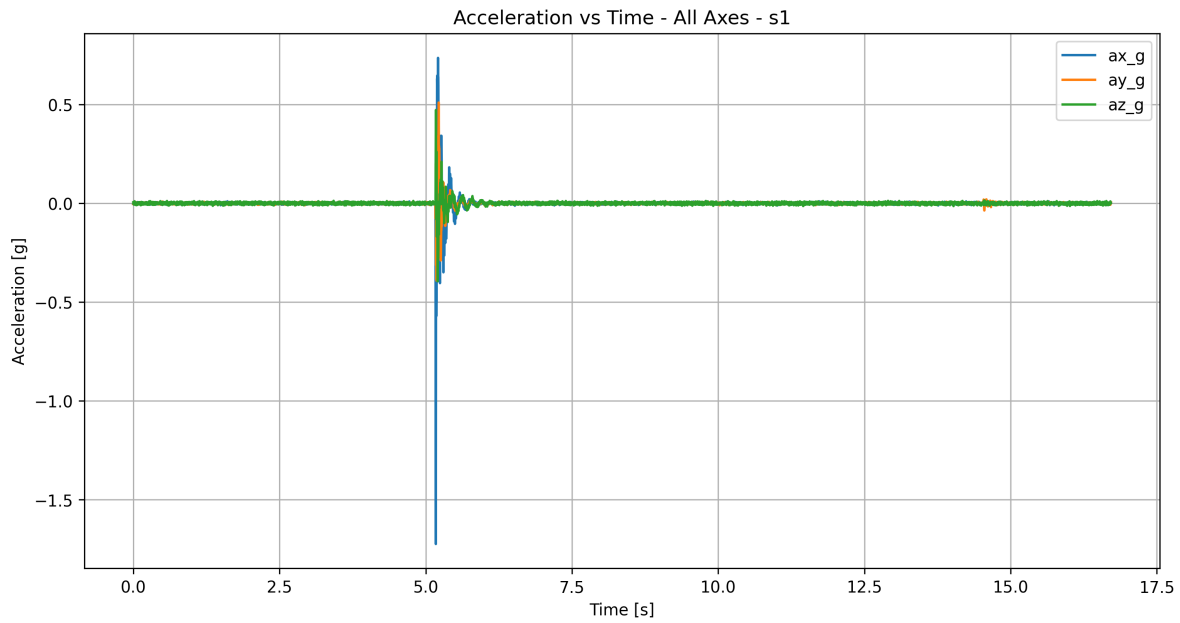


Figure 1: Acceleration vs Time for Sensor 1 mounted at the printer base.

Sensor 1 exhibited lower overall vibration amplitudes compared to the moving print head sensor. The base structure behaved relatively rigidly, with the dominant response occurring primarily in the X-axis direction.

4.2 Sensor 2 – Upper Frame

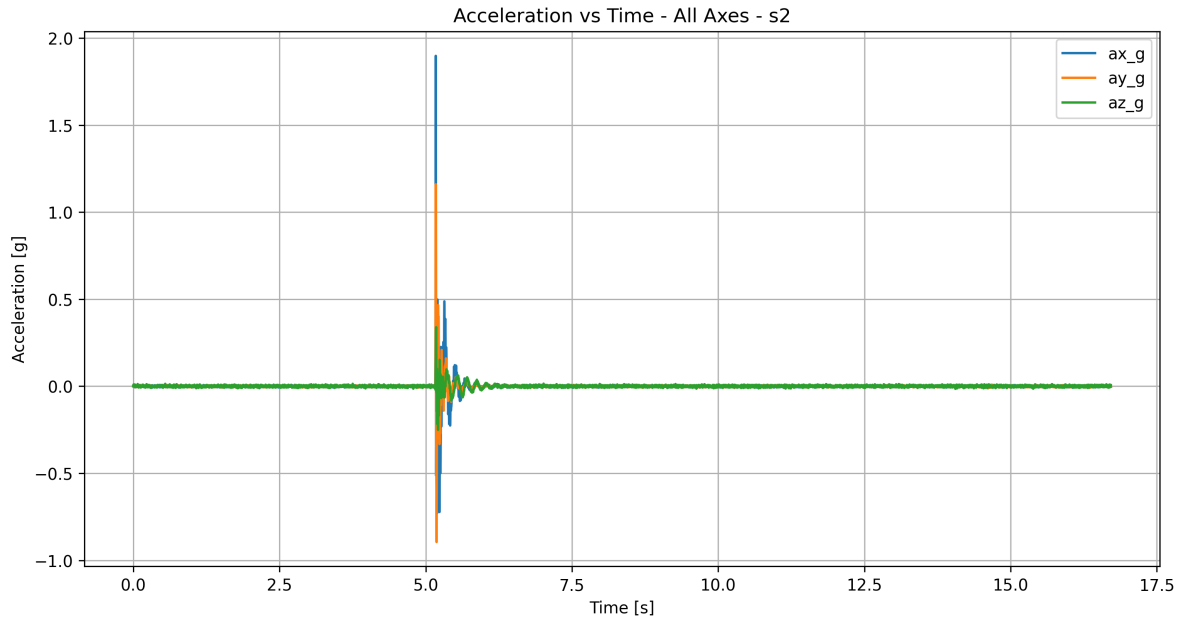


Figure 2: Acceleration vs Time for Sensor 2 mounted toward the upper section of the frame.

Sensor 2 showed increased vibration amplitudes compared to Sensor 1, indicating amplification of vibration as the distance from the base increased.

4.3 Sensor 3 – Print Head / Extruder

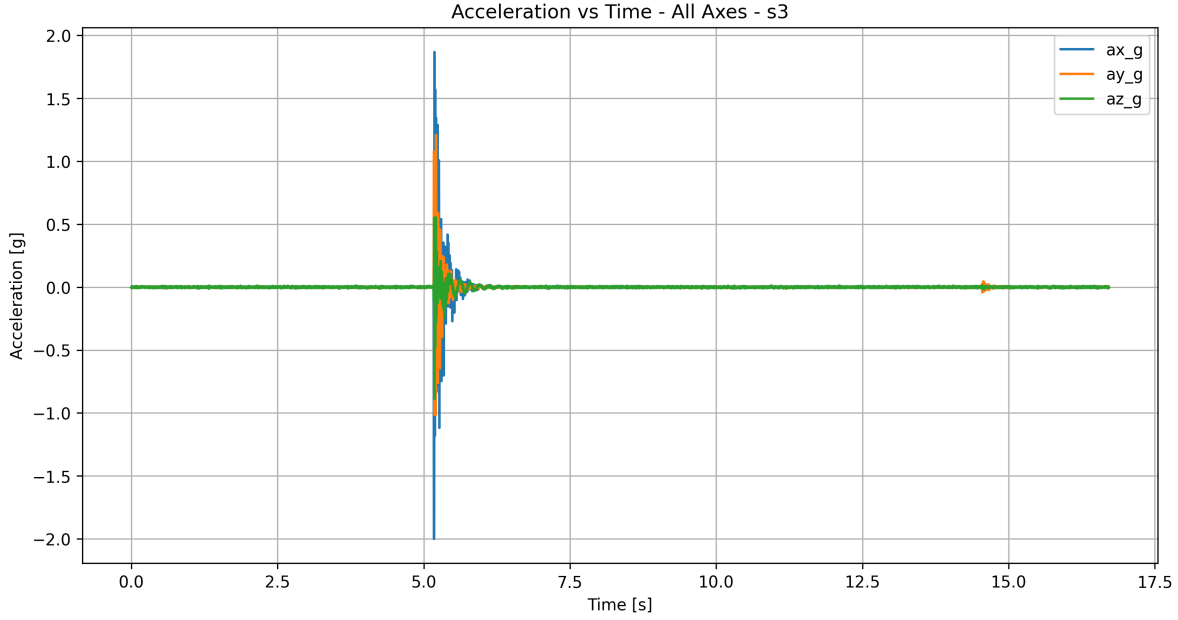


Figure 3: Acceleration vs Time for Sensor 3 mounted on the printer head/extruder assembly.

Sensor 3 produced the largest vibration amplitudes and the most energetic response overall. Since the extruder assembly is part of the moving mass of the printer, higher dynamic amplification was expected.

The print head location also exhibited a longer decay period after excitation, suggesting lower effective structural stiffness and increased dynamic compliance at that location.

5 Frequency Domain Analysis

Fast Fourier Transform (FFT) analysis was performed on the acceleration data to identify dominant vibration frequencies.

The FFT plots for each sensor are shown in Figures 4, 5, and 6.

5.1 FFT – Sensor 1

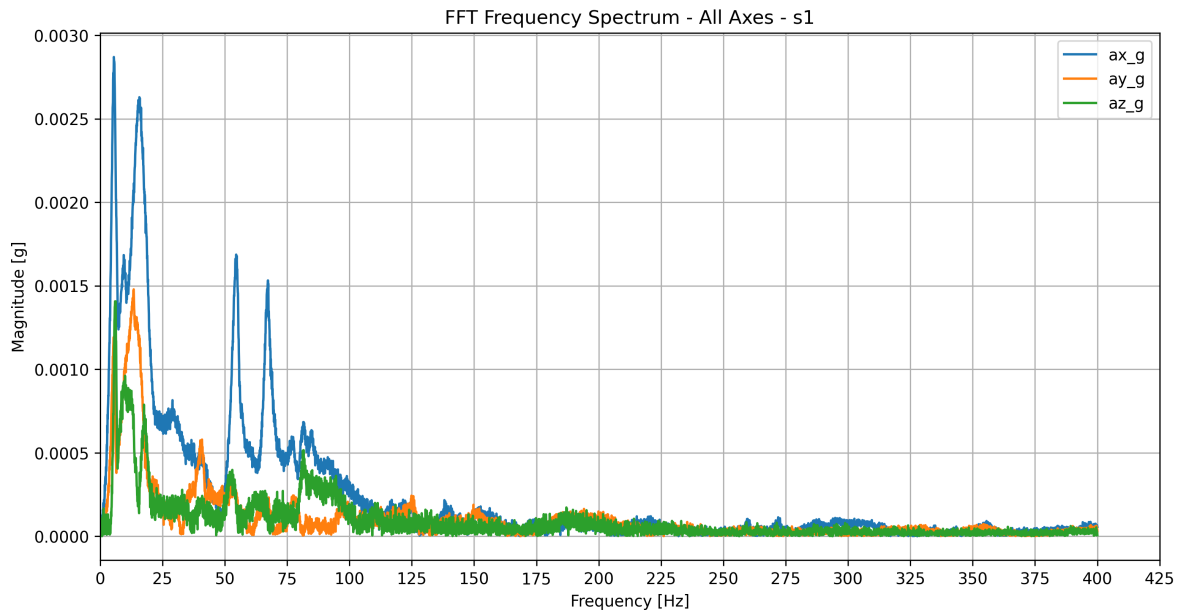


Figure 4: FFT spectrum for Sensor 1.

5.2 FFT – Sensor 2

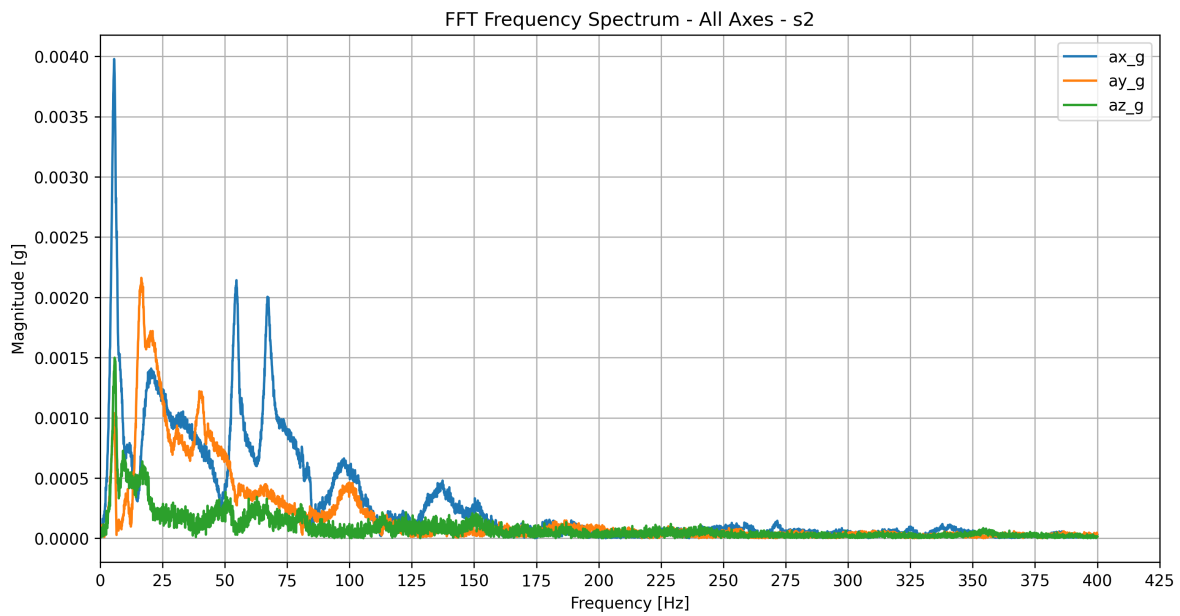


Figure 5: FFT spectrum for Sensor 2.

5.3 FFT – Sensor 3

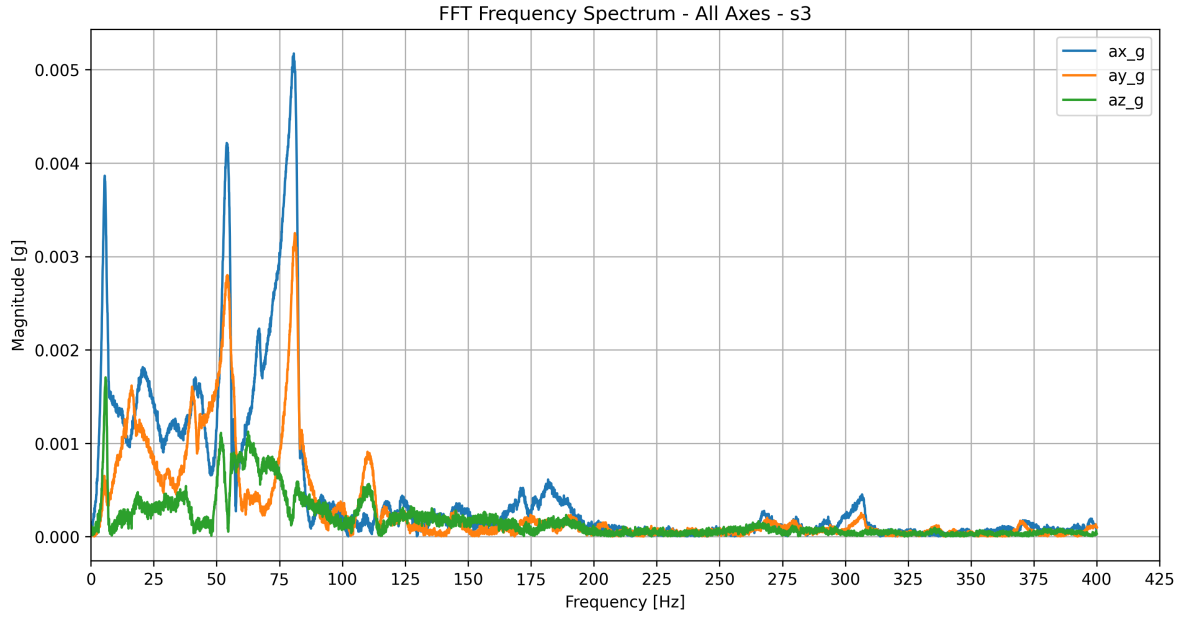


Figure 6: FFT spectrum for Sensor 3.

6 Overlapped FFT Comparison Between Sensors

To compare the structural response at each sensor location, overlapped FFT plots were generated for each axis.

6.1 X-Axis Comparison

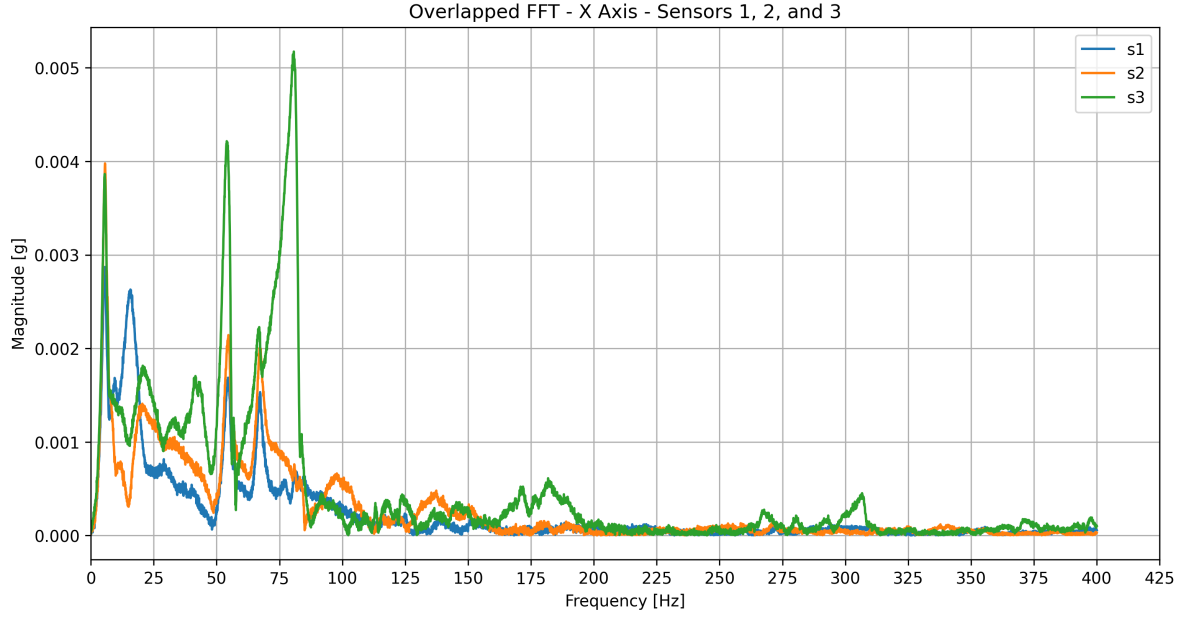


Figure 7: Overlapped FFT response for the X-axis across all sensors.

The X-axis exhibited the highest vibration amplitudes overall. Sensor 3 showed strong peaks near 80 Hz, which were significantly larger than the corresponding peaks observed at the frame-mounted sensors.

6.2 Y-Axis Comparison

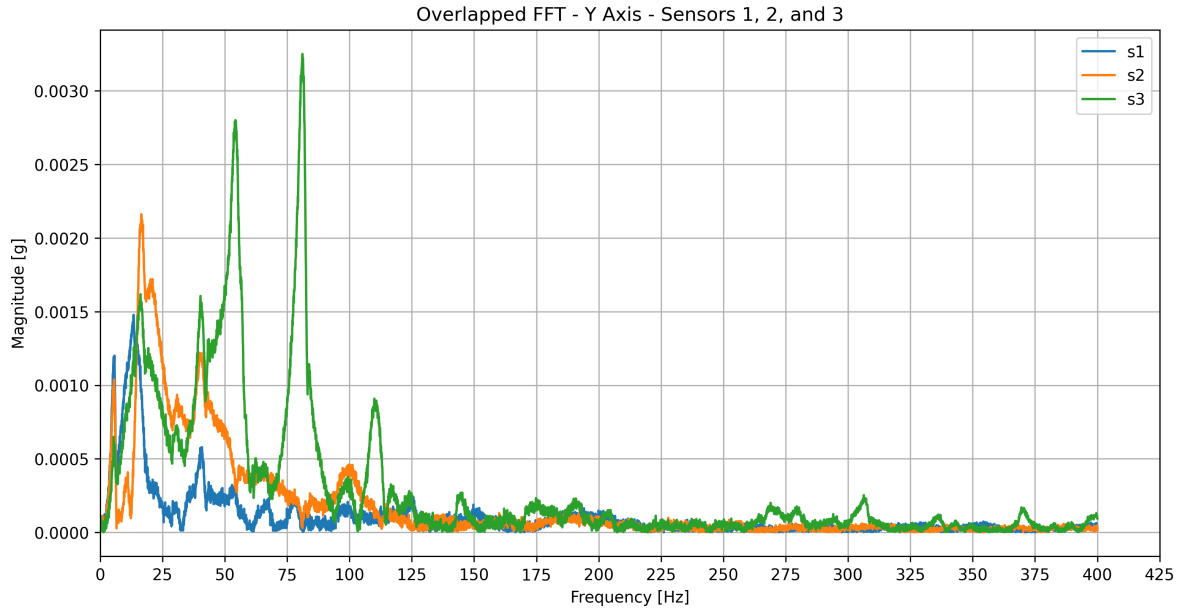


Figure 8: Overlapped FFT response for the Y-axis across all sensors.

The Y-axis response also showed strong activity in the higher-frequency range for the print head sensor. Additional peaks between approximately 50 Hz and 85 Hz were observed.

6.3 Z-Axis Comparison

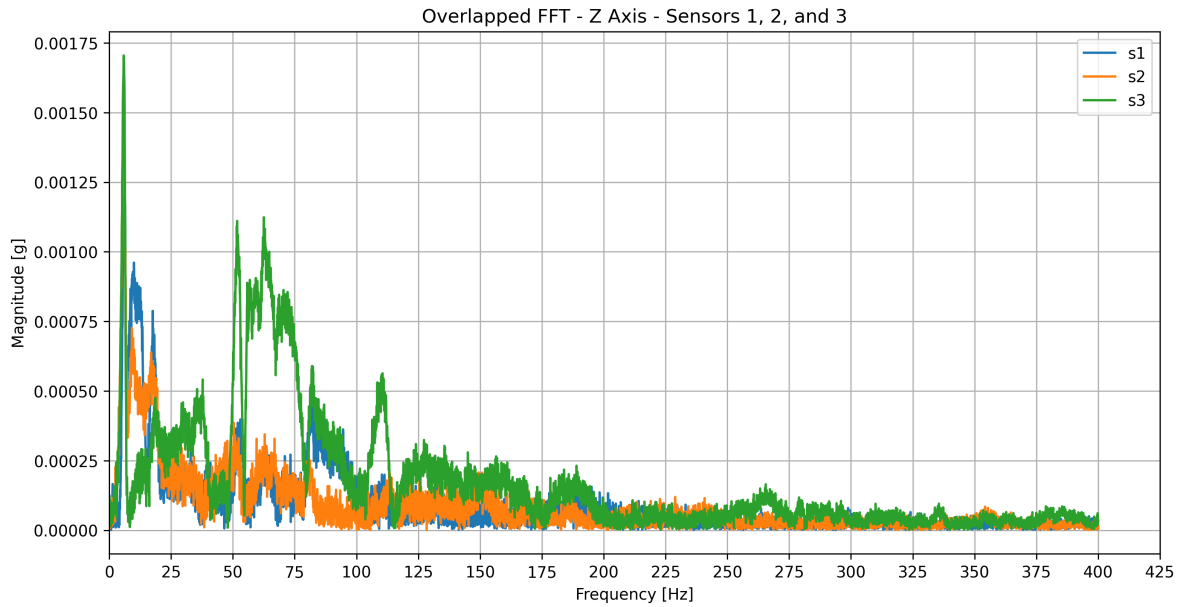


Figure 9: Overlapped FFT response for the Z-axis across all sensors.

The Z-axis exhibited lower overall amplitudes compared to the X-axis and Y-axis. However, measurable structural vibration was still present throughout the tested frequency range.

7 Dominant Frequency Results

Table 2 summarizes the dominant frequencies detected for each sensor axis.

Table 2: Dominant Frequencies by Sensor Axis

Sensor Axis	Dominant Frequency [Hz]	Dominant Magnitude [g]
s1_ax	5.50	0.002870
s1_ay	13.46	0.001479
s1_az	5.98	0.001407
s2_ax	5.62	0.003978
s2_ay	16.51	0.002162
s2_az	5.80	0.001499
s3_ax	80.71	0.005173
s3_ay	81.12	0.003250
s3_az	5.86	0.001705

8 Discussion

Several observations can be made from the current experimental results:

- The printer structure exhibits measurable vibration modes in multiple axes.
- The print head/extruder assembly experiences the largest vibration amplitudes, which is expected due to the moving mass and reduced structural rigidity compared to the base frame.
- Lower-frequency structural response near approximately 5–6 Hz was consistently observed across all sensors.
- Higher-frequency vibration components near 80 Hz were strongly present at the print head location, suggesting localized dynamic behavior associated with the moving assembly.
- The rigid sensor mounting method successfully allowed transmission of measurable structural vibration into the accelerometers.
- The use of multiple synchronized sensors provided useful information regarding vibration propagation through the printer structure.

9 Conclusions

The current experimental setup successfully demonstrated the capability of using low-cost MEMS accelerometers for multi-point structural vibration measurements.

The three-sensor configuration allowed identification of differences in vibration response between the printer base, upper frame, and print head assembly. The data indicates that the print head region experiences significantly greater dynamic excitation and contains higher-frequency vibration content compared to the frame-mounted sensors.

These results suggest that the experimental platform is capable of capturing meaningful vibration behavior and may be suitable for continued investigation into structural dynamics, resonance behavior, damping characteristics, and comparison between low-cost and higher-end sensing systems.

Future work will focus on:

- Experimental modal analysis
- Improved excitation methods
- Validation against theoretical models
- Comparison with industrial-grade accelerometers
- Investigation of resonance behavior during active printing
- Structural stiffness and damping estimation